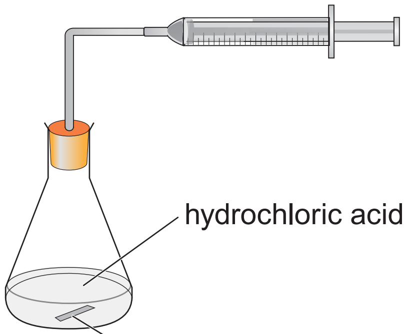
8. A group of students investigated the rate of the reaction between magnesium and dilute hydrochloric acid.

The equation for the reaction is as follows.

$$Mg \quad + \quad 2HCl \longrightarrow MgCl_2 \quad + \quad H_2$$



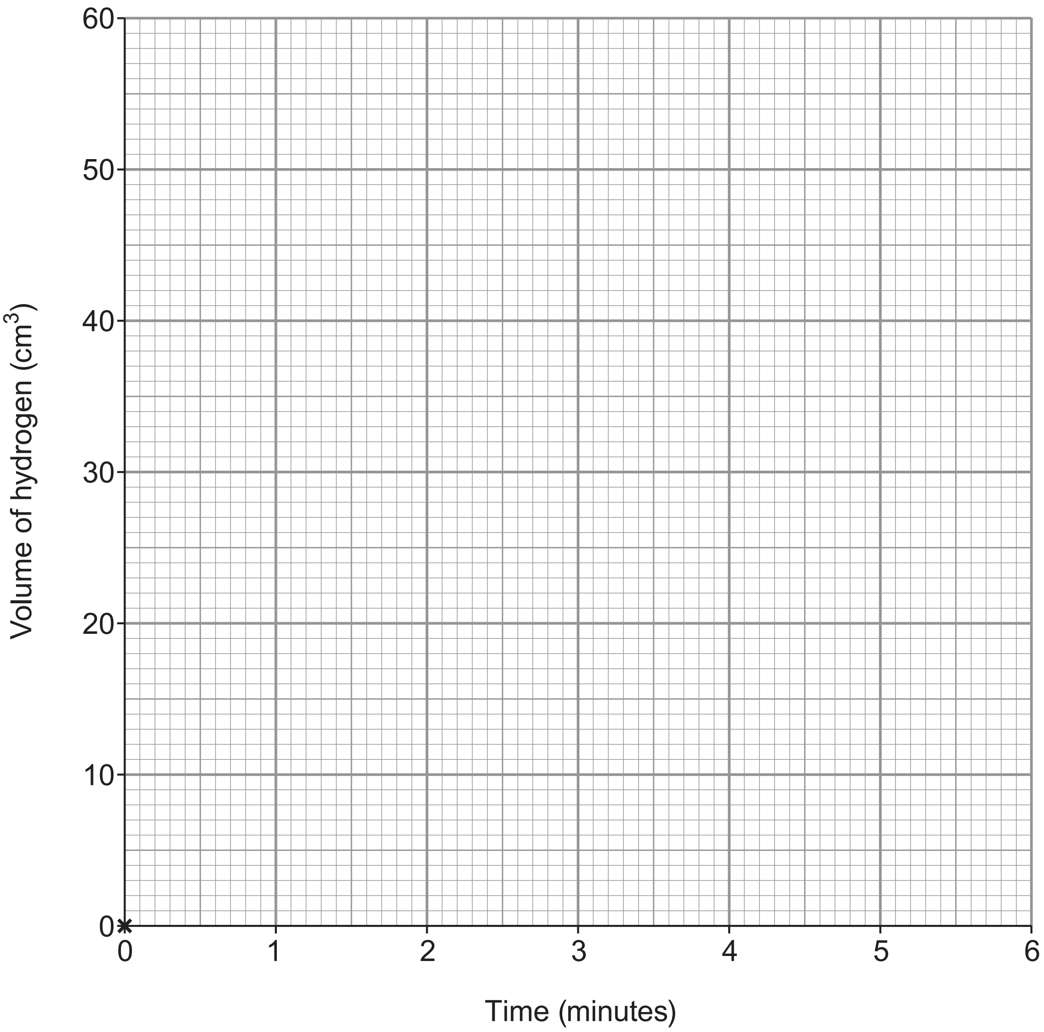
They carried out the reaction at 30 °C. The hydrogen gas was collected in a gas syringe and the volume recorded every minute for 6 minutes.

The results are shown in the table. The value at 1 minute has been left out.

Time (minutes)	0	1	2	3	4	5	6
Volume of hydrogen (cm^3)	0		29	39	46	50	50

Examiner only

(a) (i) Plot the volume of hydrogen produced against time on the grid. The first point has been plotted for you. Draw a suitable line. [3]



Examiner only

(ii) I. Use your graph to estimate the volume of hydrogen that would have been produced after 1 minute. [1]

.... cm^3

II. Calculate the mean rate of the reaction over the **first** minute. Give your answer in **cm^3/s**. [2]

Use the formula

$$\text{mean rate} = \frac{\text{volume of hydrogen } (cm^3)}{\text{time (s)}}$$

Mean rate = ... cm^3/s

(b) There is no catalyst for this reaction.

Give **two** ways the students could increase the rate of this reaction. [2]

....

....

(c) The students calculated that if they used 0.5 g of magnesium in this reaction, they would make 2.0 g of magnesium chloride. However, when they used 0.5 g of magnesium only 1.7 g of magnesium chloride was made.

Calculate the percentage yield for this reaction. [2]

Percentage yield = ... %

10

